

Lantern Ring Audio Gate MK1 - Concept Report

Date: 2026-05-28
Status: candidate concept sheet / public-safe system diagram
Surface: Lantern OS / Lantern Ring / audio gate / RAG-indexable artifact
Branch: codex/lantern-ring-audio-gate-mk1

DECISION

Converge the current Lantern Ring work into a public-safe MK1 concept artifact before any merge, production, shipment, or hardware build claim.

This report captures the system shape for review. It is not a fabrication file, electrical certification, medical/safety device claim, patent claim, order, shipment, or production approval.

CORE CONCEPT

Lantern Ring Audio Gate MK1 is a wearable control-ring concept for local-first Lantern OS interaction.

It combines:

- a 12-stone symbolic matrix;
- 3 active stones at 12, 4, and 8 o'clock;
- push-to-talk audio gating;
- Bluetooth microphone connector route;
- haptic response;
- NFC pairing cue;
- battery safety requirements;
- off-ring speaker route;
- optical self-loop validation pulse.

RING LAYOUT

The ring is modeled as a 12-position matrix. Three stones are active in MK1:

Position	Status	Intended role
12 o'clock	active	sensor return / primary gate / status marker
4 o'clock	active	input / mode / confirmation marker
8 o'clock	active	feedback / route / safety marker
Other 9 stones	passive / reserved	visual matrix, future expansion, non-active placeholders

```
flowchart TB
S12((12 o'clock\nACTIVE\nSensor return + gate))
S04((4 o'clock\nACTIVE\nMode / confirm))
S08((8 o'clock\nACTIVE\nFeedback / safety))
P01((1))
P02((2))
P03((3))
P05((5))
P06((6))
P07((7))
P09((9))
P10((10))
P11((11))

S12 --- P01 --- P02 --- P03 --- S04 --- P05 --- P06 --- P07 --- S08 --- P09 --- P10 --- P11 --- S12
```

FUNCTIONAL BLOCK DIAGRAM

```

flowchart LR
  User[Operator finger / voice intent]
  PTT[Push-to-talk gate]
  BT[Bluetooth mic connector]
  NFC[NFC pairing cue]
  Haptic[Haptic feedback]
  Battery[Battery safety gate]
  Ring[12-stone ring matrix\nactive: 12 / 4 / 8]
  Speaker[Off-ring speaker route]
  Lantern[Lantern OS local-first surface]

  User --> PTT
  PTT --> Battery
  Battery --> Ring
  Ring --> BT
  Ring --> NFC
  Ring --> Haptic
  BT --> Lantern
  Lantern --> Speaker
  Lantern --> Haptic

```

OPTICAL SELF-LOOP

The optical self-loop is a validation concept: a coded light pulse travels around the circular light pipe and returns to a sensor at 12 o'clock.

Purpose:

- confirm the ring path is intact;
- provide a non-audio local status signal;
- support a gate-ready / gate-held distinction;
- avoid claiming success from audio or Bluetooth status alone.

```

sequenceDiagram
  participant Gate as 12 o'clock gate
  participant Pipe as Circular light pipe
  participant Sensor as 12 o'clock return sensor
  participant OS as Lantern OS

  Gate->>Pipe: emit coded light pulse
  Pipe->>Sensor: pulse returns after circular path
  Sensor->>OS: report return timing / pattern
  OS-->>Gate: gate-ready, gate-held, or fault

```

ORION WATCH MK1 ENGINEERING BOUNDARY

The uploaded Orion Watch MK1 intake is now treated as the safety standard for MK1 wearable work.

MK1 should remain an existing-watch-platform prototype:

- existing round smartwatch donor platform;
- Orion / Lantern OS watchface;
- phone-side or local companion layer;
- push-to-talk as the default audio gate;
- haptic confirmation;
- wellness-only sensor language;
- custom strap, box, and presentation only;
- no custom PCB for MK1;
- no custom RF design for MK1;
- no custom battery for MK1;
- no medical claims;
- no payment NFC claim;
- no water-resistance claim beyond supplier-proven certification.

The ring concept can remain a future control-surface / docking / accessory lane, but it must not be confused with a build-ready custom electronics plan.

SUPPLIER EVIDENCE GATE

Before any sample payment or batch-of-2 lead action, request supplier evidence:

- 1. exact model number;
- 2. SoC / chipset family if available;
- 3. display type and resolution;
- 4. battery capacity and battery safety documentation;
- 5. charger specs;
- 6. FCC / CE / RoHS / UN38.3 documents if available;
- 7. Bluetooth declaration or qualification information if available;
- 8. documented water-resistance support if any IP / ATM claim is made;
- 9. watchface customization method;
- 10. app / firmware customization limits;
- 11. sample lead time and shipping method;
- 12. unit quote at 2 / 10 / 50 / 100 / 500.

PUBLIC-SAFE BOUNDARIES

Allowed in this artifact:

- concept-level ring layout;
- functional routing;
- symbolic stone positions;
- public-safe optical self-loop explanation;
- Orion Watch MK1 donor-platform requirements;
- validation gates;
- MCP/handoff status language.

Held outside this artifact:

- exact dimensions;
- electronic schematics;
- battery chemistry selection;
- enclosure tolerances;
- firmware implementation;
- manufacturing drawings;
- supplier quotes;
- certification or compliance claims;
- production, shipment, or sale claims.

BATTERY AND AUDIO SAFETY GATES

Battery safety is treated as a gate, not a solved feature. MK1 cannot be called build-ready until the following are reviewed by a qualified hardware path:

- safe charge/discharge design;
- thermal handling;
- enclosure heat risk;
- short-circuit protection;
- user-contact safety;
- transport/storage handling;
- compliance path for any real device.

For Orion Watch MK1, the safest path is factory-certified battery/charger only. Do not design a custom Li-ion pack for MK1.

Audio safety is also gated:

- push-to-talk must default to off;
- off-ring speaker route avoids forcing output into a wearable ring body;
- no hidden hot-mic claim;
- no always-on recording claim;
- local-first routing must be inspected before calling any connector live.

MCP / HANDOFF STATUS

Current status for this artifact:

```
{
  "surface": "lantern-ring-audio-gate-mk1",
  "workLoadedToMcp": "held",
  "reason": "remote repo artifact can be created, but local MCP/orchestrator queue state requires operator-machine evidence",
  "safeFallback": "branch and PR are the handoff surface until a local MCP probe records queue/tool evidence",
  "nextAction": "local agent probes MCP, verifies exposed tools, then links the PR or emits a held receipt"
}
```

VALIDATION CHECKLIST

- [x] Ring concept captured as Lantern Ring Audio Gate MK1.
- [x] 12-stone matrix included.
- [x] 3 active stones at 12 / 4 / 8 included.
- [x] Bluetooth mic connector route included.
- [x] Push-to-talk gate included.
- [x] Battery safety handled as a validation gate, not as solved hardware.
- [x] Haptic feedback included.
- [x] NFC pairing cue included.
- [x] Off-ring speaker route included.
- [x] Optical self-loop with coded pulse return to 12 o'clock sensor included.
- [x] Orion Watch MK1 donor-platform boundary included.
- [x] No custom PCB / RF / battery claim for MK1.
- [x] Wellness-only sensor language boundary included.
- [x] Supplier evidence checklist included.
- [x] Public-safe boundary included.
- [x] No merge performed by this report.
- [x] No production, shipment, certification, or fabrication claim.

RECOMMENDED NEXT ACTION

Open this branch as a PR. Use the PR as the handoff surface. Do not merge until:

- 1. remote file validation passes;
- 2. the fleet responds with status receipts;
- 3. local MCP/orchestrator evidence is either attached or explicitly held;
- 4. any hardware/safety claims remain concept-only or are moved to a qualified review lane.